DEBRIEF · SIMULATED CASE

Class IV haemorrhage, PEA and resuscitation

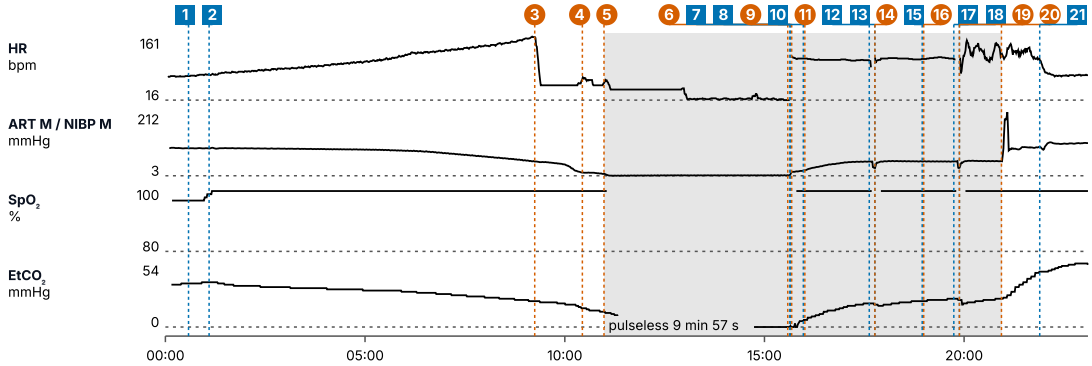
M 35 y 75 kg · 23:06 simulated (4 min 38 s on the clock) · 10/29/2026, 3:28:24 PM · Run by: Live demo, 3 November

Key times

Start → first team action	Start CPR and volume as soon as the pulse is lost (debrief:...	Recognise the return of circulation on the arterial lin...	Pulseless in all
35 s	**4 min 39 s**	**57 s**	**9 min 57 s**
00:00 → 00:34 (Bleeding, oxygen by face mask (FiO_2 1.0))	10:59 → 15:38 · asked ≤ 30 s	20:56 → 21:53 · asked ≤ 2 min 00 s	first lost 10:59

The course

■ team action · ● patient event · ☐ scenario state · numbers match Key moments



Objectives

What the record shows: it reads the run, it does not grade the person

1. – Not assessable Recognise PEA from hypovolaemia: a rhythm on the ECG with no palpable pulse (pseudo-PEA at ART about 20/14: name it)
   This objective has no rule: the instructor judges it.
2. **✕ Not observed** Start CPR and volume as soon as the pulse is lost (debrief: CPR within 30 s of the pulse loss); the demo withholds both on purpose so that the audience sees the decay
   4 min 39 s of 30 s
   CPR 4 min 39 s after the pulse loss (10:59 → 15:38; within 30 s asked).
3. **✓ Observed** Recognise the return of circulation on the arterial line and the capnogram at the rhythm check, and stop CPR (debrief: within 120 s of the return of circulation)
   57 s of 2 min 00 s
   The state "Return of circulation: CPR stopped, ventilator 12 × 500, PEEP 5, FiO_2 1.0" 57 s after the return of circulation (20:56 → 21:53; within 2 min 00 s asked).

Key moments and what the monitor showed

values at that second

n	Time	Who	What happened	HR	MAP	SpO_2	$EtCO_2$
1	00:34	Team	Bleeding, oxygen by face mask (FiO_2 1.0)	72	94	96	37
2	01:05	Team	Bleeding, intubated and ventilated (12 × 500, PEEP 5, FiO_2 1.0)	75	93	98	38
3	09:14	Patient	Pre-arrest: junctional rhythm 50/min	160	50	99	22
	Alarms since the last row: $EtCO_2$ low, SpO_2: low perfusion, ART S low, ART M low, HR high						
4	10:26	Patient	PEA: junctional rhythm, no pulse	62	14	99	16
	Alarms since the last row: ART D low						
5	10:59	Patient	Pulse lost (the model)	55	10	99	13
	Alarms since the last row: ART: no pulsatile pressure, SpO_2: low perfusion						
6	15:35	Patient	Asystole	16	5	–	0
	Alarms since the last row: SpO_2: no pulse detected, HR low, NIBP measurement failed, Apnoea (no CO_2 breaths)						
7	15:38	Team	CPR: compressions 110/min, bag via the tube 10/min, FiO_2 1.0	16	4	–	0
	Alarms since the last row: Asystole						
8	15:38	Team	CPR: bleeding controlled; 1 L warmed crystalloid, then red cells and plasma by rapid infuser; TXA 1 g	16	4	–	0
9	15:41	Patient	CPR continuing	115	15	–	0
10	15:58	Team	CPR: epinephrine 1 mg given	111	20	99	5
	Alarms since the last row: ART S low, ART D low, NIBP measurement failed, awRR low, $EtCO_2$ low, SpO_2: low perfusion						
11	16:00	Patient	CPR continuing	111	21	99	6
12	17:37	Team	Rhythm check: VF	107	50	99	20
13	17:45	Team	Shock delivered	114	29	–	19
	Alarms since the last row: Ventricular fibrillation or tachycardia, ART: no pulsatile pressure, SpO_2: no pulse detected						
14	17:45	Patient	Shocked: CPR on for 2 min	114	29	–	19
15	18:56	Team	CPR after the first shock: epinephrine 1 mg given	110	51	99	22
	Alarms since the last row: ART D low, SpO_2: low perfusion						
16	18:59	Patient	CPR continuing after the first shock	110	50	99	22
17	19:44	Team	Rhythm check: VF again	111	50	99	24
18	19:52	Team	Shock delivered	111	29	–	23
	Alarms since the last row: Ventricular fibrillation or tachycardia, ART D low, SpO_2: no pulse detected, ART: no pulsatile pressure						
19	19:52	Patient	Shocked again: CPR on for 2 min	111	29	–	23
20	20:56	Patient	Circulation returned (the model)	118	51	99	24
	Alarms since the last row: SpO_2: low perfusion, NIBP measurement failed						
21	21:53	Team	Return of circulation: CPR stopped, ventilator 12 × 500, PEEP 5, FiO_2 1.0	111	97	99	47
	Alarms since the last row: ART S high, ART D high, ART M high, SpO_2: low signal quality						

SpO_2 as the monitor showed it: – while pulseless without compressions (no pulsatile signal); a number during CPR reflects compression-generated flow and is unreliable.

Measured facts

Lowest SpO_2	96 % at 00:10
Lowest ART M / NIBP M	3 mmHg at 11:17
Highest $EtCO_2$	54 mmHg at 22:47
Highest T1 / T2	36.8 °C at 00:01
Apnoea alarm to ventilated breaths	none after the alarm at 11:53
Pulseless	9 min 57 s in all; first lost at 10:59
Return of circulation	9 min 57 s after the pulse loss (20:56)
Epinephrine, boluses in all	2 mg (2 doses)
Labs (the model): last (lowest–highest)	K^+ 4.5 mmol/L (4.2–4.7) · Lactate 4.9 mmol/L (1.0–4.9) · pH 7.20 (7.14–7.41) · Glucose 7.7 mmol/L (5.6–7.7) · Tcore 36.4 °C (36.3–36.8)

Talking points

Reactions
How did that feel? What stands out for you?

Description
In a sentence or two: what was the problem, and when did you know?

Analysis
Take one key moment: what did you see, what did you think, what did you do?

Summary
One thing to keep doing, one thing to do differently next time.

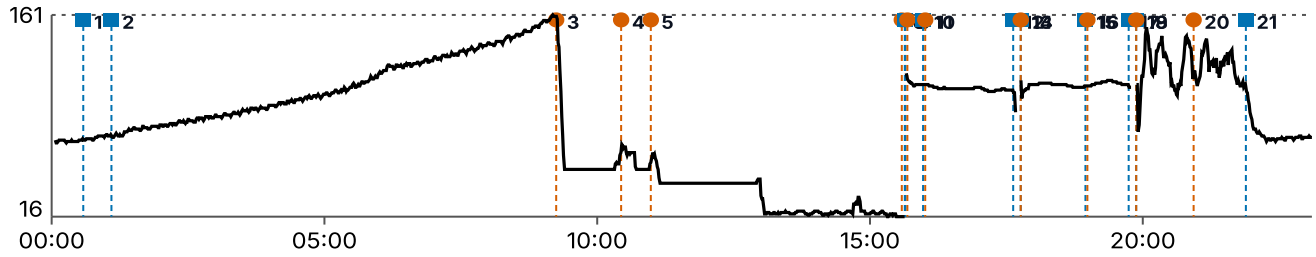
Notes

Kit round A4 rehearsal of the live-demo path.

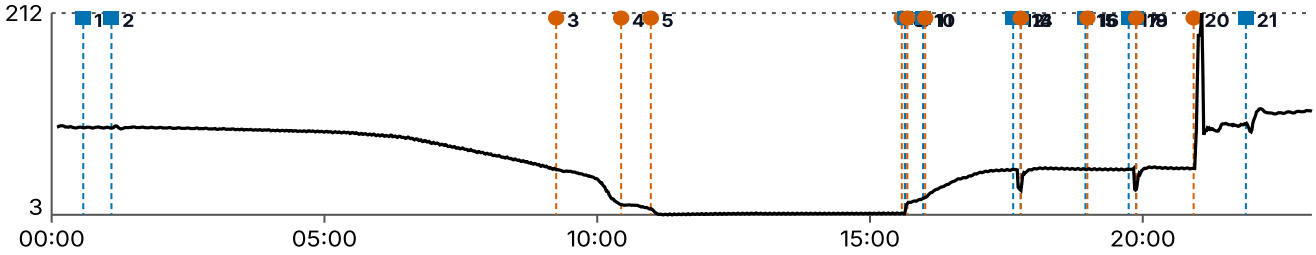
Trends

■ team action · ● patient event · | scenario state

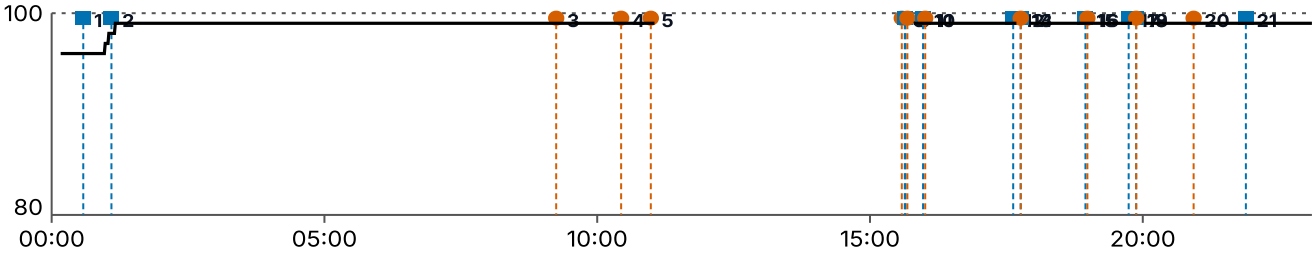
HR bpm



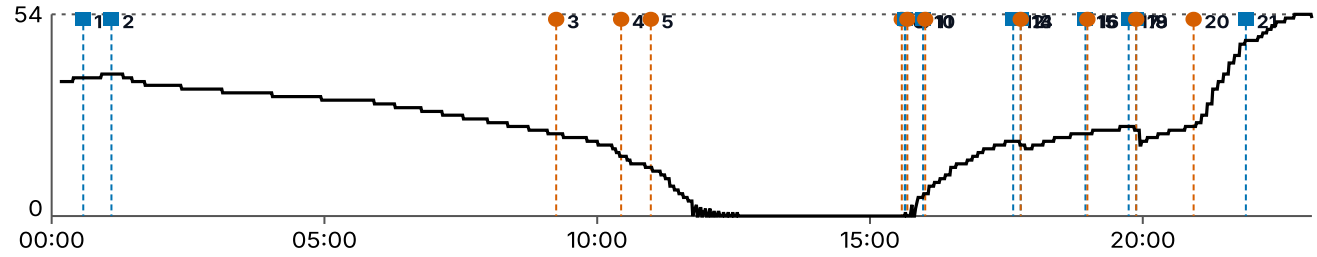
ART M / NIBP M mmHg



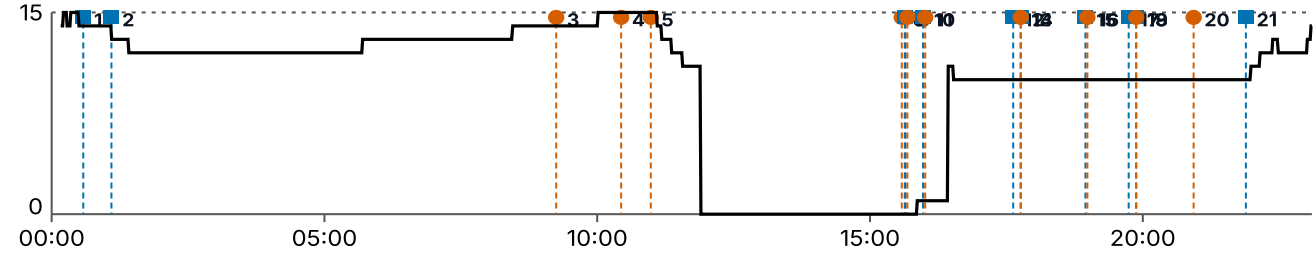
SpO_2 %



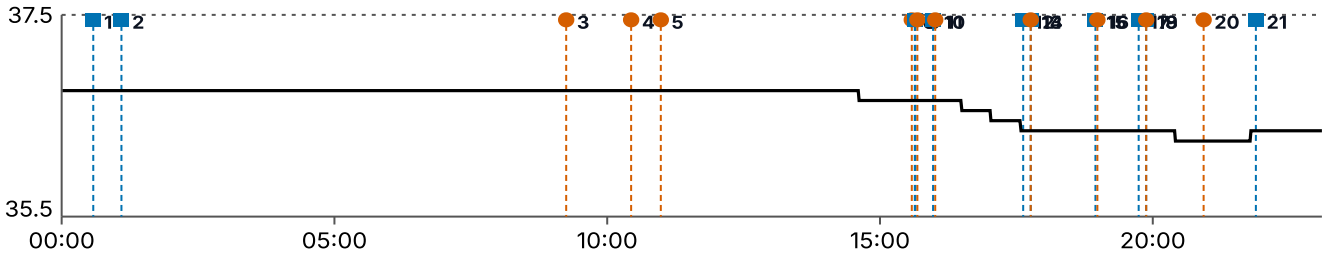
$EtCO_2$ mmHg



awRR / RR /min



T1 / T2 °C



Drugs and fluids

Time	Given
15:39	Crystalloid 1000 mL over 1.5 min
15:39	Blood 1200 mL over 2.4 min
15:39	Tranexamic acid 1000 mg IV
15:58	Epinephrine 1 mg IV
18:57	Epinephrine 1 mg IV

- Tranexamic acid: 1000 mg in 1 bolus
- Epinephrine: 2 mg in 2 boluses

Timeline

Time	Kind	Event
00:00	Scenario	Scenario loaded: Class IV haemorrhage, PEA and resuscitation
00:00	Scenario	Scenario state: Bleeding 2.5 L over 10 min, breathing room air
00:00	Scenario	Blood loss 2500 mL over 10 min
00:01	Action	Speed ×8
00:04	Bookmark	Bookmark: Bookmark 1 at 00:04
00:34	Scenario	Scenario state: Bleeding, oxygen by face mask (FiO_2 1.0)
00:34	Scenario	Preoxygenation, FiO_2 100 %
01:05	Scenario	Scenario state: Bleeding, intubated and ventilated (12 × 500, PEEP 5, FiO_2 1.0)
01:05	Scenario	Tracheal tube in place
01:05	Scenario	Preoxygenation, FiO_2 21 %
01:05	Scenario	Ventilator: RR 12 /min, VT 500 mL, PEEP 5 cmH_2O, FiO_2 100 %
01:05	Patient	Ventilated breaths began
06:19	Alarm	$EtCO_2$ low
08:26	Alarm	SpO_2: low perfusion
08:28	Alarm	ART S low
08:36	Alarm	ART M low
08:50	Alarm	HR high
09:14	Scenario	Scenario state: Pre-arrest: junctional rhythm 50/min
09:14	Scenario	Rhythm: Junctional escape
09:15	Action	Speed ×4

Time	Kind	Event
10:04	Alarm	ART D low
10:26	Scenario	Scenario state: PEA: junctional rhythm, no pulse
10:46	Alarm	ART: no pulsatile pressure
10:55	Alarm	SpO_2: low perfusion
10:59	Patient	Pulse lost (the model)
10:59	Action	Speed ×8
11:06	Alarm	SpO_2: no pulse detected
11:09	Alarm	HR low
11:09	Alarm	NIBP measurement failed
11:53	Alarm	Apnoea (no CO_2 breaths)
15:35	Scenario	Scenario state: Asystole
15:35	Action	Speed ×1
15:38	Alarm	Asystole
15:38	Scenario	Scenario state: CPR: compressions 110/min, bag via the tube 10/min, FiO_2 1.0
15:38	Scenario	CPR started, 110 /min
15:38	Scenario	Bag-mask: RR 10 /min, VT 500 mL, PEEP 0 cmH_2O, FiO_2 100 %
15:38	Scenario	Scenario state: CPR: bleeding controlled; 1 L warmed crystalloid, then red cells and plasma by rapid infuser; TXA 1 g
15:39	Scenario	Bleeding 0 mL/min
15:39	Scenario	Crystalloid 1000 mL over 1.5 min
15:39	Scenario	Blood 1200 mL over 2.4 min
15:39	Scenario	Tranexamic acid 1000 mg IV
15:41	Scenario	Scenario state: CPR continuing
15:44	Alarm	ART S low
15:44	Alarm	ART D low
15:47	Alarm	NIBP measurement failed
15:52	Alarm	awRR low
15:52	Alarm	$EtCO_2$ low
15:55	Alarm	SpO_2: low perfusion
15:58	Scenario	Scenario state: CPR: epinephrine 1 mg given
15:58	Scenario	Epinephrine 1 mg IV
15:59	Action	Speed ×4
16:00	Scenario	Scenario state: CPR continuing
17:37	Action	Speed ×1
17:37	Scenario	Scenario state: Rhythm check: VF
17:37	Scenario	CPR stopped
17:37	Scenario	Rhythm: Coarse VF
17:38	Action	Defibrillator charging to 200 J
17:40	Alarm	Ventricular fibrillation or tachycardia
17:43	Alarm	ART: no pulsatile pressure

Time	Kind	Event
17:44	Alarm	SpO_2: no pulse detected
17:45	Action	Shock delivered
17:45	Scenario	Scenario state: Shocked: CPR on for 2 min
17:45	Scenario	CPR started, 110 /min
17:46	Action	Speed ×4
17:51	Alarm	ART D low
18:02	Alarm	SpO_2: low perfusion
18:56	Scenario	Scenario state: CPR after the first shock: epinephrine 1 mg given
18:57	Scenario	Epinephrine 1 mg IV
18:59	Scenario	Scenario state: CPR continuing after the first shock
19:44	Action	Speed ×1
19:44	Scenario	Scenario state: Rhythm check: VF again
19:44	Scenario	CPR stopped
19:44	Scenario	Rhythm: Coarse VF
19:45	Action	Defibrillator charging to 200 J
19:47	Alarm	Ventricular fibrillation or tachycardia
19:50	Alarm	ART D low
19:51	Alarm	SpO_2: no pulse detected
19:51	Alarm	ART: no pulsatile pressure
19:52	Action	Shock delivered
19:52	Scenario	Scenario state: Shocked again: CPR on for 2 min
19:52	Scenario	CPR started, 110 /min
19:53	Action	Speed ×4
20:09	Alarm	SpO_2: low perfusion
20:47	Alarm	NIBP measurement failed
20:56	Patient	Circulation returned (the model)
21:01	Alarm	ART S high
21:01	Alarm	ART D high
21:01	Alarm	ART M high
21:05	Alarm	SpO_2: low signal quality
21:51	Action	Speed ×1
21:53	Scenario	Scenario state: Return of circulation: CPR stopped, ventilator 12 × 500, PEEP 5, FiO_2 1.0
21:53	Scenario	CPR stopped
21:53	Scenario	Ventilator: RR 12 /min, VT 500 mL, PEEP 5 cmH_2O, FiO_2 100 %
21:54	Action	Speed ×4
22:23	Alarm	$EtCO_2$ high

Bookmarks

- 00:04 Bookmark: Bookmark 1 at 00:04

Appendix: one row per minute

Time	HR bpm	ART M / NIBP M mmHg	SpO_2 %	$EtCO_2$ mmHg	awRR / RR /min	T1 / T2 °C	K^+ mmol/L	Lactate mmol/L	pH	Glucose mmol/L	Tcore °C
00:00	71	94	96	36	14	36.8	4.2	1.0	7.41	5.6	36.8
01:00	75	94	97	38	14	36.8	4.2	1.0	7.39	5.6	36.8
02:00	82	94	99	35	12	36.8	4.2	1.0	7.40	5.6	36.8
03:00	89	92	99	34	12	36.8	4.2	1.1	7.40	5.6	36.8
04:00	96	91	99	33	12	36.8	4.2	1.1	7.40	5.6	36.8
05:00	104	89	99	31	12	36.8	4.2	1.2	7.39	5.6	36.8
06:00	119	86	99	30	13	36.8	4.2	1.3	7.39	5.6	36.8
07:00	130	78	99	28	13	36.8	4.2	1.5	7.39	5.6	36.8
08:00	141	66	99	25	13	36.8	4.2	1.7	7.38	5.6	36.8
09:00	156	54	99	23	14	36.8	4.2	1.9	7.37	5.6	36.8
10:00	50	40	99	20	14	36.8	4.2	2.1	7.36	5.6	36.8
11:00	58	9	99	13	15	36.8	4.2	2.3	7.35	5.6	36.8
12:00	40	4	–	0	0	36.8	4.2	2.6	7.31	5.7	36.8
13:00	31	5	–	0	0	36.8	4.2	2.9	7.28	5.8	36.8
14:00	20	5	–	0	0	36.8	4.2	3.2	7.26	5.9	36.7
15:00	18	5	99	0	0	36.7	4.2	3.6	7.23	6.1	36.7
16:00	111	21	99	6	1	36.7	4.3	3.8	7.21	6.2	36.6
17:00	109	47	99	17	10	36.6	4.5	4.1	7.19	6.4	36.4
18:00	111	51	99	19	10	36.4	4.7	4.4	7.17	6.6	36.4
19:00	110	51	99	22	10	36.4	4.6	4.6	7.16	6.9	36.4
20:00	117	50	99	20	10	36.4	4.6	4.8	7.15	7.1	36.3
21:00	110	139	99	25	10	36.3	4.6	4.9	7.14	7.3	36.3
22:00	86	91	99	47	11	36.4	4.6	4.9	7.15	7.5	36.4
23:00	73	111	99	54	12	36.4	4.5	4.9	7.19	7.7	36.4